2) $\left(\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{array}\right)$

RREF: $\left(\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{array}\right)$

a) If x_3, x_5 is param:

$$\Rightarrow \begin{cases} x_4 = 2 - x_5 \\ x_2 = x_5 - 2 \\ x_1 = 1 - x_5 - x_3 \end{cases}$$

b) ~~Yeah~~ If we take x_1, x_2 as param:

$$\Rightarrow \begin{cases} x_5 = -2 - x_2 \\ x_4 = 2 - x_5 = 4 + x_2 \\ x_3 = 1 - x_5 - x_1 = 3 + x_2 - x_1 \end{cases}$$

Still ok

c) ~~If we take x_2, x_5~~

~~$\Rightarrow \begin{cases} x_4 = 2 - x_5 \end{cases}$~~

Because $x_2 - x_5 = -2$, they are bounded and cannot be parameterized.

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3) $\begin{pmatrix} a & 0 & b & 2 \\ a & a & 4 & 4 \\ 0 & a & 2 & b \end{pmatrix}$

REF: $\begin{pmatrix} a & 0 & b & 2 \\ 0 & a & 4-b & 2 \\ 0 & 0 & b-2 & b-2 \end{pmatrix}$

RREF: $\begin{pmatrix} a & 0 & 0 & 2-b \\ 0 & a & 0 & b-2 \\ 0 & 0 & b-2 & b-2 \end{pmatrix}$

$$\Rightarrow \begin{cases} z = \frac{b-2}{b-2} \\ y = \frac{b-2}{a} \\ x = \frac{2-b}{a} \end{cases}$$

One soln: $\begin{cases} a \neq 0 \\ b \neq 2 \end{cases}$

No soln: $\begin{cases} b \neq 2 \\ a = 0 \end{cases}$

Inf soln with 1 paran: $\begin{cases} a \neq 0 \\ b = 2 \end{cases}$

Inf soln with 2 paran: $\begin{cases} a = 0 \\ b = 2 \end{cases}$

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c) $\begin{cases} x + 2y + 3z = 4 \\ 3x - 4y + z = 2 \\ x + y - 5z = 3 \end{cases}$

$$\left(\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 3 & -4 & 1 & 2 \\ 1 & 1 & -5 & 3 \end{array}\right) \xrightarrow{R_3 - R_1} \left(\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 3 & -4 & 1 & 2 \\ 0 & -1 & -8 & -1 \end{array}\right) \xrightarrow{R_2 - 3R_1} \left(\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 0 & -10 & -8 & -10 \\ 0 & -1 & -8 & -1 \end{array}\right)$$

$$\xrightarrow{R_2 - R_3 - 10R_1} \left(\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 0 & 0 & 72 & 0 \\ 0 & -1 & -8 & -1 \end{array}\right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 0 & -1 & -8 & -1 \\ 0 & 0 & 72 & 0 \end{array}\right)$$

$$\xrightarrow[R_3 = \frac{1}{72} R_3]{R_2 = -R_2} \left(\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 0 & 1 & 8 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow{R_2 - 8R_3} \left(\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\xrightarrow{R_1 - 2R_2 - 3R_3} \left(\begin{array}{cccc} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

. So $\begin{cases} x = 2 \\ y = 1 \\ z = 0 \end{cases}$

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